PROJECT GUIDE

Orbi

An AirPods-style connection popup for Windows 11

Earbuds, Wi-Fi, Ethernet and USB drives. A small popup slides up the moment something connects, then goes away. Orbi opens with Windows and tells you what is already connected.

MADE BY
Gopinath

VERSION
0.1.0

PLATFORM
Windows 11

BUILT WITH
Tauri 2 + Rust

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1.4 MB
installer

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SECTION 01

Why Orbi exists

A MacBook shows a card the moment AirPods connect. Windows shows a short sound and a small toast that is easy to miss. Orbi gives Windows one clear popup for any kind of connection, at almost no cost in CPU or memory.

One glance

See which earbuds connected and the battery of each bud and the case, without opening Settings.

Any connection

Earbuds, Wi-Fi, Ethernet and USB drives share one popup, with a layout for each.

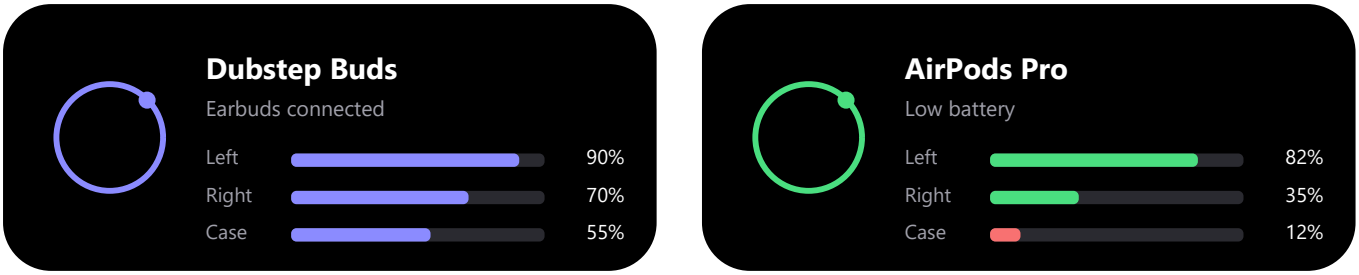
Out of the way

The popup shows for about 5 seconds and is then destroyed, so nothing sits on screen.

Local and light

No account, no cloud, no polling. Windows tells Orbi when something changes.

How it looks



Drawn illustrations of the popup. The real popup is a 380 by 270 card at the bottom centre of the screen with the device artwork, battery bars and a close button.

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SECTION 02

Every feature

Everything below ships in this repository today.

Devices

- Bluetooth earbuds: battery from Windows, with the full pair and case drawn when only one value is reported.
- AirPods: left, right and case battery read from Apple's Bluetooth advertisements (unit tested, not yet confirmed on real AirPods).
- Wi-Fi: network name, signal strength and link speed.
- Ethernet: connected adapter and link speed.
- USB drives: drive letter, how full it is and free space (removable drives only).

Behaviour

- Popup on connect, and again when a battery newly drops to 20% or below.
- A device that reconnects within 2 seconds does not pop up twice (connection flapping).
- Hover over the popup to keep it open; use the close button to dismiss it.
- The tray tooltip lists every connected device and its battery.
- Per-type switches in the tray menu, remembered between runs.

On launch

- Starts with Windows by default on first run. You can turn it off in the tray menu.
- When Orbi opens, it checks what is already connected and shows one popup for the most useful device: earbuds, then Wi-Fi, Ethernet, USB.
- After that, every new connection shows a popup automatically.

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SECTION 03

How it looks: every layout

The layout is chosen from the device type. All share the Orbit design: a ring with a glowing dot.

Type	Shows	Source
Earbuds	Name, left and right bud, case, battery bars, charging mark	Windows Bluetooth battery value
AirPods	Same as earbuds with real left, right and case values	Bluetooth advertisement data
Wi-Fi	Network name, signal bars, speed	Windows WLAN API
Ethernet	Adapter name, link speed	Windows network interface table
USB drive	Drive letter, used and free space	Windows volume notifications

Artwork for the buds and case is drawn as SVG (src/art.ts) with a lookup by device name, so a known product can get its own look.

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SECTION 04

Tech stack

Small on purpose: a native shell with a web page used only for the popup.

Piece	Choice	Why
Shell	Tauri 2 (Rust)	Tiny installer, low idle memory, tray support
Popup UI	TypeScript, Vite, SVG	Fast to draw, one small page
Windows APIs	windows crate 0.62	Direct access to Bluetooth, network and volume notifications
Settings	JSON file	Simple and easy to inspect
Autostart	tauri-plugin-autostart	Registers Orbi to start with Windows
Installer	NSIS	About 1.4 MB

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SECTION 05

Architecture

Watchers push events into one core. The core decides what is worth showing.



Each watcher subscribes to a Windows notification and sleeps until it fires. Then it reads the device again, compares with the last read and sends only what changed as a normalized DeviceEvent. One consumer thread owns the core. It applies your switches, debounces flapping, and tells the popup module to show or the tray to update its tooltip.

Why this shape

- No polling: idle CPU is near zero because every thread is blocked waiting on Windows.
- One event type means the core and the popup never care which watcher spoke.
- The core is plain Rust with no Windows calls, so it is fully unit tested.
- The popup window is created on demand and destroyed after it closes, so idle memory is only the tray process.

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SECTION 06

How each part works

One module per job in src-tauri/src.

File	Job
lib.rs	Builds the tray menu, applies first-run autostart, starts the pipeline.
events.rs	The normalized DeviceEvent: id, type, name, status, battery parts, metrics.
core.rs	Live-device state, 2 s debounce, 20% low-battery rule, launch popup choice.
pipeline.rs	Connects watchers to core to popup. Holds launch events for 1.5 s and shows one popup.
popup.rs	Creates the popup window, closes it after about 5 s, honours hover and close.
settings.rs	Reads and writes settings.json, per-type switches, popup time.
watchers/bluetooth.rs	Connected Bluetooth devices and battery; ignores switched-off paired devices.
watchers/airpods.rs	Parses Apple advertisements into left, right and case levels.
watchers/network.rs	Wi-Fi and Ethernet via network interface change notifications.
watchers/usb.rs	Removable drives via volume interface notifications.
mock.rs	Fake devices for testing without hardware.

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SECTION 07

Repository map

Top level of github.com/goprocker/Orbi.

Path	What it is
src-tauri/	Rust app: tray, watchers, core, popup window, installer config
src/	Popup UI: main.ts (layouts), art.ts (SVG artwork), styles.css
index.html	Popup page entry
app-icon.png	Source for the Orbi icon set
Orbi-Tech-Plan.pdf	Plan, architecture and progress log
Orbi-Guide.pdf	This guide
README.md	Short status, requirements and run steps

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SECTION 08

Install and run

Windows 11 only.

Requirements to build

- Rust (stable) and the Visual Studio C++ Build Tools (Desktop development with C++).
- Node.js and npm.
- WebView2 (included with Windows 11).
- Windows Smart App Control must be off, because it blocks unsigned Rust build helpers.

Run from source

```
npm install
```

```
npm run tauri build -- --debug --no-bundle
```

```
src-tauri\target\debug\orbi.exe --mock=mock_airpods
```

Use npm run tauri build, not a plain cargo build. A plain cargo build makes the popup load the dev server and show a blank error page.

Install

Run npm run tauri build. The installer is written to src-tauri/target/release/bundle/nsis/Orbi_0.1.0_x64-setup.exe. Run it and Orbi starts in the tray.

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SECTION 09

Using Orbi

There is no main window. Orbi lives in the tray.

First launch

- Orbi starts and adds itself to Windows startup (first run only).
- It checks what is already connected and shows one popup within about two seconds.
- From then on, connecting earbuds, joining a network, plugging in Ethernet or inserting a USB drive shows a popup.

Tray menu

- Check items for Earbuds and Bluetooth, Wi-Fi, Ethernet and USB drives.
- Start with Windows.
- Mock entries to preview each popup without hardware.
- Quit Orbi.

The tooltip

Hover the tray icon to see every connected device with its battery, or 'Orbi: no devices'.

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SECTION 10

Settings

Stored in %APPDATA%\com.orbi.app\settings.json. Missing or broken values fall back to defaults.

Setting	Default	Meaning
bluetooth	true	Show popups for earbuds and Bluetooth devices
wifi	true	Show popups for Wi-Fi
ethernet	true	Show popups for Ethernet
usb	true	Show popups for USB drives
popup_seconds	5	How long the popup stays (clamped to 2 to 30)
autostart_initialized	false	Set after the first run so turning autostart off sticks

Disabled types still pass disconnect events, so Orbi's live state stays correct when you switch them back on.

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SECTION 11

Build, test, debug

Tests

cd src-tauri, then cargo test. 24 unit tests cover the core rules, settings, AirPods parsing, network and Bluetooth logic and the launch popup choice.

Debug without hardware

Start with --mock=<id> or use the tray Mock entries. Ids: mock_earbuds, mock_airpods, mock_dubstep, mock_wifi, mock_ethernet, mock_usb.

Logs

A debug build prints lines starting with [orbi] to the console, for example which watchers started and any popup or autostart error.

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SECTION 12

Privacy, troubleshooting, uninstall

Privacy

Orbi reads device names, battery levels, the network name and drive space on your PC only to draw the popup. It sends nothing anywhere and has no account.

Troubleshooting

Problem	Try
Popup does not appear	Check the tray switch for that device type is ticked.
Build is blocked	Turn Windows Smart App Control off.
Blank popup in a dev build	Use npm run tauri build instead of a plain cargo build.
Wrong earbuds shown	Orbi ignores paired devices that are switched off. Reconnect the earbuds.
AirPods left and right swapped	Known risk until verified on real AirPods.

Uninstall

Uninstall Orbi from Windows Settings, Apps. Settings live in %APPDATA%\com.orbi.app and can be deleted by hand.

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SECTION 13

Hard problems and limits

What was difficult and what is still open.

Solved

- Windows keeps the last battery value of switched-off paired earbuds. Orbi only trusts values from devices Windows reports as connected.
- WebView2 normally uses about 175 MB. Tuned launch flags bring the popup helper to about 82 MB in a debug build test.
- Mica or acrylic was rejected: it fills the rectangular window and ignores the card's rounded corners.

Known limits

- USB hard drives and SSDs are not detected, because Windows reports them as fixed disks.
- Bluetooth and AirPods code needs real hardware to verify fully.
- The release build with the tuned flags has not been re-measured.
- At launch only one popup is shown, even if several devices are connected.

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SECTION 14

Words you need

Word	Meaning
Tray	The icons by the clock where Orbi lives.
Watcher	A module that listens to Windows for one kind of change.
Event	A small record: which device, what happened, battery and metrics.
Debounce	Ignoring a repeat that comes too soon after the last one.
WebView2	The browser engine Windows provides to draw the popup page.
Tauri	A framework for desktop apps with a Rust core and a web UI.
Autostart	Starting a program automatically when Windows starts.

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SECTION 15

Cheat sheet

Want to	Do this
Run from source	npm install; npm run tauri build -- --debug --no-bundle
Preview a popup	orbi.exe --mock=mock_airpods, or use the tray Mock entries
Run the tests	cd src-tauri; cargo test
Make the installer	npm run tauri build
Stop it starting with Windows	Untick Start with Windows in the tray menu
Change how long it shows	Edit popup_seconds in settings.json

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